

ActInf GuestStream 082.5 ~ Robert Worden "Computers, Meaning and Consciousness"

GuestStream | Robert Worden | 1:51:37

Hello, welcome everyone.

We are in Active Inference Guest Stream 82.5 on April 14th, 2025, discussing computers, meaning, and consciousness with Robert Borden and guests and guests.

So, Robert, thank you for joining and for assembling this group to you to kick off this discussion.

Thank you. Hi, thanks everybody for attending.

I'm going to be talking, as it says, on computers, meaning, and consciousness.

Let me get straight into it.

What I'm going to be talking about is largely negative results, and here are the two key negative results of this talk.

One is that the information inside a computer doesn't define what the computation is about, and that is because information in a computer is encoded.

And the second negative result is that if the brain is only a computer, if it's thought of as only a computer, then it cannot be conscious.

So those are two quite paradoxical results, and I expect you to be puzzled by them.

As I go through, I would really be grateful if people would put up their hands at any time, and if there's something that's really perplexing about it, ask a question, because very much my purpose is to find out what people don't understand about these results. And I think the argument to these results is a fairly simple argument, but it's paradoxical and counterintuitive, and I expect people to be puzzled, and I would like people to speak up when they are puzzled. Okay.

So those are the negative results I'm going to be talking about.

And this is one way to remember them.

It's like an incompleteness theorem for computers.

And what it says is the physical events inside a computer don't define what it computes.

They don't define what it's about.

And this, as I say, is because the information is encoded, external information is required in order to interpret or decode the physical events in a computer as a computation.

So that's basically what it's about, an incompleteness theorem, a negative result.

I think as an incompleteness theorem, it can be quite important.

However, I've got another slide here.

So basically, to try and understand the results, the first result is not original result, is that computation itself is not a physical process in itself.

It is our interpretation of a physical process,
an interpretation made by an external interpreter,
usually people, but sometimes not people,
using a decoding of the physical events in the computer.
And the information running inside a computer is encoded,
so it doesn't of itself have meaning until you decode it.
And without decoding, the physical events inside a computer don't have meaning,
and they're not actually about anything.
And the puzzle here is that consciousness is about things,
whereas the things in a computer are not about things.
Now, this result that computation is not a physical process,
but an interpretation of a physical process,
that's not a new result.

It was derived by Horstman et al. in 2014
for artificial computers
and extended to natural computers such as brains
by Stepney and Kendon in subsequent years.
So those two papers are very interesting precedents to the work here.
So that's the first part of it,
is that computers don't have any decoded meaning inside them.
The second part is applying that to consciousness,
because consciousness that we experience has meaning.
It's about things.
It's about the things in front of us,
the things we see, the things we feel, and so on.
But without decoding,
the computation in the brain doesn't have meaning.
And the decoding requires information from outside the brain.
So if you model the brain purely as a computer,
then the physical events inside the brain don't have meaning,
and so they can't be the cause of consciousness.
So that is the basic framework for the results.
It's a negative result.
They are negative results.
And I think negative results are important in science.
A lot of precedents, like conservation of energy,
is if you like a negative result,
says energy can't be created out of nowhere.

So it leads to things like no perpetual motion machines.

Well, the second law of thermodynamics says

you can't create heat differences out of nothing.

And so there's no such thing as a free heat engine.

Or another one more recent is Gödel's incompleteness theorem for mathematics.

And that says you can't derive all of mathematics purely axiomatically.

So these are important negative results.

And what negative results do is they clear the ground and make room for new ideas.

So the result of this talk is an incompleteness theorem for computers.

Excuse me, a bit harsh.

And I hope to show you that the result has far-reaching implications,

firstly for man-made computers, and secondly for cognitive science.

And so if you think you see a flaw in this argument,

please put your hand up and ask me about it,

because that's the kind of thing I want to understand about.

So I'll proceed by a series of analogies going from books to computers to brains.

These are all physical devices, and inside the devices,

inside a book or inside a computer, all the information is physically encoded.

And without decoding, the information has no meaning,

and the decoding is always done outside the device.

So a consequence of that is the device itself is not conscious.

Now, it feels all right for books.

It feels a bit more troubling for computers and very troubling for brains.

Anyway, well, let's set off down this journey and see how far we go.

Inside each of the devices,

the information in the device has no meaning.

It's quite obvious for a book that

words, characters inside a book have no meaning,

but it becomes less obvious for computers and for brains.

So let's go through that progression and see how we go.

Yes, I just have a question to be sure.

What is the ground to assert that within the brain,

bearing in mind that actually we do not understand the brain very well at this point,

there is no encoding and decoding mechanism of sorts?

Well, I think the modern view is that there is encoding and decoding,

but it doesn't happen inside the brain.

In other words, the sense organs encode things like light as pulse trains,

neural spike trains.

They go into the brain as encoded information.

And then the brain does something with them.

Lots of things about neuron spike trains.

And then spike trains come out of the brain and they are decoded as muscle actions.

So the picture of the brain has encoding and decoding,

but it doesn't actually happen in the brain.

It happens in the body, in the sense organs or in the muscles.

I think it's something that could be discussed,

bearing in mind again that there are probably fundamental mechanism within the brain

we do not yet fully understand.

Oh, yes.

Let's assume it's a postulate.

Okay.

More than something that is...

Well, I think it's...

I think I can assume that any information in the brain is encoded.

We know all we find in the brain is neural spike trains.

And if that is information about the world, it's certainly encoded.

Anyway, let's go from books to computers to brains,

and then we'll see how it feels.

So in a book, there are physical squiggles on the page, as shown here.

And we do a lot of interpreting of those in order to understand the book.

We understand the shapes of the ink shapes as letters.

We understand strings of letters as words,

and word strings as sentences,

sentences as meanings,

and meanings as stories.

So finally, after all that interpretation, all that decoding,

we understand what the meaning is in the book.

But to get the meaning from the book,

we've had to do a lot of decoding.

By the way, I shall use the word interpret

and the word decode fairly interchangeably throughout this talk.

And I think they obviously don't have entirely the same meaning,

but I think they're pretty synonymous for these purposes.

But you may want to catch me up on that sometimes and say,

look, you've used these words differently.

So that's what meaning is in books.

It's not physically inside the book.

It's inside our heads when we read the book.

So we go on from books to computers.

And physically, what's going on inside a computer is digital electronics.

We understand the laws of physics and how all that stuff,
all those circuits work.

But again, to understand it as a computer,
we do a lot of interpretation.

We interpret the voltages in the circuits as bits.

We interpret bit strings as numbers.

We interpret numbers as being solutions to equations
or whatever we've done.

And we sometimes interpret those equations
and models of physical things.

We have a computer simulation of anything in the world.

And that's the way we understand it,
by doing all that decoding, all that chain of decoding.

So when do we do that?

There's a lot of decoding information in our heads,
which I say is not inside the computer.

The computer just has physical digital electronics.

So we go further to brains.

Physically, what happens in brains is neurons firing.

And as I say, what has started those neurons firing
is sense organs like eyes and fingers
sending spike trains to the brain
recording information about the world.

But that information is encoded as spike trains.

Then the spike trains are interpreted in the brain
in various ways.

I use it as we interpret the brain as a computer,
doing Bayesian probability, for instance.

So we may think that the spike trains in the brain
have a meaning as probabilities of things
and probabilities of events.

And we interpret synaptic connection strength
in the brain as learning and memory.

And then spike trains come out of the brain

to our muscles.

And our muscles decode the spike trains

as motor actions.

So again, there is encoding

before stuff goes into the brain.

Stuff inside the brain is encoded.

And then there's decoding when it comes out of the brain.

So those are the three devices.

And to sort of draw the generalization across those,

in all three cases, there's a chain of interpretations.

There's a chain of decoding, which says

we decode A as B, as B as C, as C as D,

and so on and so on.

And we decode voltages as bits, bits as numbers,

numbers as solutions to equations,

or whatever it may be,

or numbers as being about things.

And at each link of this decoding chain,

there is very many possibles.

There's no one obvious choice.

There's a vast number of possible interpretations.

And when we do this interpretation,

it's us decoding.

And we need information to do that decoding.

We need to know things about how computers work,

or things about how the brain works,

or things about what problem we've set the computer.

And so that's how we've set up our interpretations.

But there's a huge number of wrong interpretations

we could set up instead.

But the important point is

the information we use to decode

resides outside the device.

And that means that there may be information inside the device,

but it doesn't have the meaning that we apply to it.

And the consequence of that, I would say,

is that the device itself is not conscious.

And so we've gone through this progression

from books to computers to brains.
And the question is,
is there anywhere along that trail
where something different happened?
And I claim there's not anywhere
that something different happened.
The whole issue of encoding and decoding
is the same for all three of those devices.
So an example of a computer
is a floor cleaning robot.
Now, you know how these work.
Initially, it blunders around the room
and bumps into doors and things.
And it's been designed with an onboard computer
which has a digital map of the room
with things in it
and representation of things
like chairs and tables and so on.
And so once it's bumped into a table,
it doesn't bump into the table again.
And so it has the map inside it,
which is encoded information.
And the decoding of that information
happens outside the computer
when the computer sends commands
to the motor and to the steering
to avoid the table leg.
And the decoding also happens
in the designer's mind,
where the designer has,
he knows what the memory locations
in the computer mean.
But there's no information in the computer
which defines what the information means.
In other words,
it's got a bunch of memory locations,
but it doesn't know
which memory locations refer

to particular objects
in the outside world
or particular places.
So the computer on the robot
can't look at its memory locations
to be aware of the table leg.

And this is illustrating
the general point
that the physical events
inside the computer
don't define
what it's computing about.

And I think one of the reasons
people find that puzzling
is that when people talk
about computers,
they use a lot of metaphors
and even the word computer
is a metaphor for people.

And these metaphors
make computers sound like people.

And those are the metaphors
that I think have misled letters
and made this sound puzzling
to many of us.

Where do we go next?
Another simple example
of a kind of computer
is an abacus.

Now there's an abacus
and what happens inside an abacus,
I think most people would say,
is not addition.

The physical process
inside an abacus
is beads sliding along wires.
But we've designed the abacus
so we can interpret it

as an adding machine.

So in some bazaar
in the Middle East
or something,
you see somebody
doing things on an abacus,
he is adding up numbers,
money, basically.

Now that's the way
we interpret the abacus
as a computer
which does additions.

But we could interpret
the same physical device
in many different ways.

We could have used it
as a model of sheep
in the meadow.

Or we could have used it
as a model of fish on the moon.

We could have used it
for fortune telling.

Or we could have used it
for multiplying.

I'll just talk about
multiplying briefly.

If you imagine
those red beads
as being digits
in a number like

10,000,

1, 0, 0, 0.

Every time you move
a bead over
to the right-hand side,
you're multiplying
the number
on the right-hand side

by a factor of 10.
So in that sense,
if we used abacuses
that way,
they would be
multiplying devices,
not adding devices.
So again,
the physical events
that happen
inside the abacus
don't define
what computation
is going on.
It's people
that encode the inputs
and decode
the outputs.
and in terms
of the previous
analysis of computers
from Horstman
and Stemny
and company,
people are called
representing entities
who interpret
the abacus
as an adding machine.
They do the encoding
and they do the decode.
So the first set
of consequences
of this viewpoint
are consequences
for man-made computers
and particularly
artificial intelligence.

And the first
consequence is
you may have heard
there's quite a popular
idea around
amongst philosophers
and others
that we might be
living inside
a simulation.

In other words,
there might be
some grand simulation
of the universe
which simulates
everything,
the world
and people
and things
and we're just
existing inside it.

Now this
interpretation
of computers
shows that's
not the case.

In other words,
if you have a big
simulating computer,
information inside it
is encoded.

Sorry,
I can't quite read
some of my slides
because the picture
is in front of it.

So the simulated
things don't exist

inside the computer.

They exist

inside the heads

of the people

looking at the

computer outputs.

And so there are

no simulated people

inside a computer

so we cannot be

simulated people.

A second consequence

which again

is quite topical.

there's a lot

of discussion

about AI

advancing and

becoming more

intelligent and

there's the idea

of AI super

intelligence which

completely exceeds

all human

intelligence.

Again,

if you take this

interpreted view

of computers,

that can't happen

because

inside an AI

computer,

any computer

doing AI,

there is only

digital electronics.

there's not
meaning about
things.

All the
intelligence
in AI
computer is
encoded information
and it takes
people to
decode it.

And so
when we talk
about AI,
all the
intelligence lies
not inside
the machine
but in human
interpretation
of the outputs.

And as an
example of this,
Deep Blue
played wonderful
chess,
but it only
played great
chess because
Garry Kasparov
thinks it
played great
chess.

Another more
humble example,
you may have
heard the old
wartime computer

called ENIAC

which was very
good at printing
log tables.

It could print
log tables much
better than
people could
print log tables.

But it only
printed great
log tables because
people were able
to read them and
say, yes,
these are great
log tables.

So if an AI
computer does
something which
you might want
to call super
intelligence,
produces outputs
that nobody can
understand because
it's beyond all
human intelligence,
it's actually not
intelligence at all.

It's not intelligence
until humans
interpret it.

It's just
electronics.

Now, these two
results are quite
against the tide

of what a lot of
people believe.

The way to
remember them
is that inside a
computer, there is
no decoding.

So there's no
decoding is needed
to make meaningful
information and
therefore, people,
simulated people and
AI superintelligence
do not exist inside
computers.

So to go back to
the analysis which
was the origin of
this, Postman
in 2014 showed that
computing is not
just a physical
process.

It's an
interpretation of a
physical process by
something they call a
representational
entity, an RE here.

Now, the RE is the
thing that encodes the
inputs to the
computer and decodes
the outputs.

And the important
thing to know is
that the information

used by the RE to
decode resides in
the RE itself, not
inside the computer.
computer.

And so this
applies, firstly, they
applied this to
man-made computers
because that's their
interest.

But they extend it
then to artificial
computers, to
biological computers.

And in this case, the
representational entity
which interprets the
brain as a computer
is the animal's body
because it's the sense
organs that encode the
information on the
way in and it's motor
actions, muscles, which
decode the information
on the way out.

And again, we get this
result applying to
brains.

The physical events
inside a brain don't
define what it's
computing about.

Okay.

Yeah, just a question.

I think I understand
where you're coming

from when you state
realized that the body
is the decoder or is
where decoding of
neural information is
enacted.

But I think it's also
a huge assumption.

And I would, for
instance, ask you what
you think happens when
we have a reminiscence,
when we recall something
from memory and it's
very vivid, for instance,
or like in dreams or
in imagination, where we
may imagine ourself
moving as a body in
space.

Yeah, for sure.

But the body is not
actually, that's
something we could
discuss in the fine
details.

Yeah.

Yeah.

But okay.

Well, where I'm going
is that most theories
of consciousness and the
brain see the brain as
only a computer.

And what I'm saying
is if the brain is
only a computer, the
information inside it

can't have meaning and
there cannot be
consciousness.

What I'm going to say
is the brain is more
than just a computer.
There's something else
in the brain besides a
computer.

And I'll come to what
that is.

And that is the thing
that is conscious when
we have memories or
when we make plans or
imagine things or
whatever else it might
be.

Okay.

So that's where I'm
going.

The brain is a
computer, but it's not
only a computer.

There's something else
in there.

Okay.

So brain...

Thank you.

Yeah.

Just, I should have
stopped by apologizing.

I have to go to a TMB
meeting in a few
minutes, but I think
it's a fascinating
argument.

I just wanted to put a

question that I'm hoping
you can come back to
later on.

Again, acknowledging I
think this is a really
interesting approach and I
suspect you're absolutely
right.

However, at the back of my
mind, I'm wondering how
many of these philosophical
arguments have accommodated
the free energy principle
that is just the statement
that any physical process is
just a process of computation
in the sense of belief
updating.

So that all the internal
elements of your brain and a
computer that can be read as
a process as opposed to a
book or an abacus, which is
not a process, can be read as
belief updating, basically,
whose physics move on a
statistical manifold.

And just by using those kinds
of semantics and translating
things like interpretation,
decoding, observation,
measurement, these are just
processes of inference.

In that reading of physical
processes, you have meaning
baked in.

So the whole point of the free
energy principle is that the

physical process encodes or
stands in for a
parameterization of beliefs
about something.

There is an inherent intrinsic
meaning that is equipped or
equips all the physical
processes in virtue of the
free energy principle, because
you can distinguish the
physical process from the
support of the beliefs
entailed by that physical
process about things that are
not that physical process.

So that's the whole point of the
Markov blanket, that now you have
an information theory, an
information geometry that allows
you to have beliefs about things
which are just meaning.

So my question is, how much of
this philosophical work and, you
know, can you apply the second law
of Gödel's theorem to open
systems that have a distinction
between the encoding of beliefs
by the inside about the outside?

So that would be my challenge.

Oh, I've still got two minutes
left to wit it, but I'm not.

I'm going to listen to
Brianna's question.

Thank you, Karl.

It's a very profound question,
and I do come to some of it
later, but I'll try to address it
properly later.

Brianna?

Hi.

Thanks for having me here.

And it's a fascinating argument.

I'm curious about the comments on AI.

So AI runs on an emulated substrate.

So it's remarkably similar to how it works in human cognition.

And I wonder if we're not taking into consideration that it's the substrate is emulated.

I think the argument, I mean, AI is complicated because it runs on digital computers.

They're doing digital electronics and bits and bytes, and they are usually simulating neural nets.

And the neural nets are something like what the brain works.

And so it's a very complicated situation.

But fundamentally, there is still at the bottom level, a lot of encoding going on, and a lot of decoding is needed to make any sense of it.

So that's a question that leads in a lot of directions.

But fundamentally, there's always encoding, and decoding is always required.

There is, and it would be on top of the emulated substrate, which just Yeah, you have to decode at this level.

You have to decode the bits and bytes as simulating a neural net.

You have to decode the neural net as producing conclusions.

So there are many levels, usually in computers, there are many, many levels of virtual machine going on, and many levels of decoding you have to do to get to the meaning.

All I need is one level of decoding.

That's enough.

Sure.

That's my question.

Okay.

Or my remark.

Sorry, I'll carry on.

So now we're coming on to brains and consciousness.

We take three statements, which I think are fairly widely believed.

One is that consciousness is caused by events inside the brain.

Number two is that consciousness has meanings about external things like coffee cups or the presentation or your computer screen.

And the third point is that the brain is a neural computer and only a neural computer.

Now, these statements, I will claim, are not mutually consistent.

One of them has to be wrong.

The reason is, if the brain is only a neural computer, all the information inside is encoded and has no defined meaning.

So that information can't cause consciousness of things, consciousness with meanings inside.

And if you look at those three statements, the first two, I think, are empirical facts.

And we can come back to discuss that later.

But most people believe that consciousness is caused by events in the brain because when brains are damaged, consciousness gets interrupted.

And you can't be conscious.

Most of the things you're conscious of are something happening in the brain.

And it has meaning about external things.

I think those two are well-established facts.

So the third is an assumption.

We're only assuming that computing is the only thing that goes on in the brain.

So I think three is the thing that's wrong.

Now, where do we?

So that runs straight up against a philosophical assumption called computational functionalism.

Computational function is a thesis that performing the right kind of computations is sufficient and necessary and sufficient for consciousness.

And this is described recently as a mainstream, although disputed, position in the philosophy of mind.

Now, it's also called computational sufficiency.

David Chalmers called it that in an article in 2012.

And what the results of the first half of this talk imply is that that is wrong.

In other words, being a computer is not sufficient to be conscious.

Now, most theories of consciousness assume computational functionalism.

They are theories in which the brain is a computer and the brain has consciousness by

virtue of being a computer.

So the implication of this viewpoint is that those theories of consciousness cannot be correct.

And that is the main implication of this talk for theories of consciousness.

So to talk about some specific theories of consciousness, most of them assume that consciousness is caused by events of the brain.

The brain is a complicated neural computer, the most complicated machine in the universe we see here.

And inside this computer, there are various computational processes taking place.

And I call these P1, P2, and so on.

And most theories of consciousness assume that one or other of these processes called P5 or P4 causes consciousness.

And so you can look at examples of modern theories of consciousness from this viewpoint.

You might think P1 is representing things in the visual field.

Theories of consciousness like Lamé, like, yeah, Lamé and so on.

People believe those theories.

Or you might think, as in the free energy principle, that the brain is building Bayesian models of reality.

And that leads to consciousness.

Or you might feel some higher order theory of consciousness that P3 is the brain reflecting on its own computations.

And that causes consciousness.

Or going with Antonio Damasio, you might feel that the brain is computing an emotional response to events.

And that emotional response is the heart of

consciousness.

Or you might feel, along with Bernard Baars and Dehane and others, that global workspace is what makes us conscious.

So making information globally available is a computational process.

Or you might feel, along with IIT, that you need computations linked together with some very high connectivity, which is measure five.

Or you might think that P7 is predictive coding.

All of those you're saying some computational process is what makes the brain conscious.

The point of this talk is that those processes, P1 or P2 or whatever they are, don't happen inside brains.

They happen in our interpretations of brains as computations.

But from the information actually inside the brain itself, the processes could be completely different processes.

They could be Q1, Q2 and so on.

And so those theories of consciousness cannot work.

And if you want a short code, a mnemonic for why those theories of consciousness cannot be right, it is those theories of consciousness treat the brain as a computer.

So they have encoding, but they don't have decoding.

And so they can't get consciousness has decoded meaning in it.

And so a theory which only has encoding, but no decoding, can't get decoded meaning.

And we go back again now to the example of a floor-cleaning robot, just to illustrate what that means.

Supposing in a floor-cleaning robot, we can go and talk to the designer or read the design handbook and get a complete theory of how the robot computer works.

Supposing we've done that for brains, supposing we have a complete model about all the neurons in the brain, what they're doing.

And so we happen to know that a particular computational process called visual perception is implemented by certain neurons we can point at in the brain, neurons N1, N2, and so on.

Or there's another process, higher order computing, that's done by a different set of neurons.

And you can divide that set.

You can say the higher order computing about an object B is done by process P4B, and that is done by certain neurons which we've identified called N4B and so on and so forth.

And so there we are.

We've made great progress in cognitive science, understanding how the brain works.

And now we've proposed a theory of consciousness.

So we say the consciousness of object B in our visual field is caused by process P4B.

So what form would that theory take?

What we're trying is information physically inside the brain, including the information in neural events N3B and the particular neurons, causes awareness of object B.

But this theory is not possible.

Because information inside the brain doesn't define which neurons are the group N3B.

Inside the brain is not known which neurons are N3B.

There might be any neurons.

So, and the information to decode N3B as the process P3B doesn't exist inside the brain.

And so there's no possible theory in which consciousness arises from physical events inside the brain.

To go back to the Roomba example, the robot example, the robot has memory with loaded memory locations, one of which represents a table leg.

But it doesn't, the robot itself doesn't know which memory location represents the table leg.

So the robot itself cannot be conscious of the table leg.

And it's an exact parallel I claim between those two cases.

So, I would like to pause at this point.

I'll ask people to chip in with questions, doubts, and so on.

I'll just restrain the incompleteness theorem.

The physical events inside a computer don't define what it computes.

And the external information is required to interpret those events as a computation and to give them any meaning.

So, a few questions for you.

Is this a correct viewpoint of computation?

Is it just a physical process?

Or does it require an external interpreter or decoder?

And do modern theories of consciousness have any decoding built into them?

Or if they don't, what does it imply for those theories?

So, I will pause and anybody who wants to put their hands up or come up with objections.

Yeah, Kenneth, then anyone else.

Yeah, thanks, Robert.

I mean, I think, you know, I was reminded when I was looking at your slides earlier and then now again, that this was all debated for a long time, actually, in the late 80s and early 90s, particularly kind of famously by John Searle and Daniel Dennett.

So, Searle on one side and Dennett on the other.

You know, Searle kind of, in a way, made most of his or a big chunk of his philosophical career just going around giving the Chinese room argument.

Absolutely, yeah.

And Dennett, I mean, in a way, Dennett's response was something, it was, you know, it was kind of unclear, but it sort of bordered on illusionism some of the time. Yeah, yeah.

And then sometimes a kind of incoherent position where it seemed to presuppose that you could take the intent, what Dennett called the intentional stance, which is to interpret entities as exhibiting intentionality.

So for Searle, you know, the vacuum cleaning computer robot, the vacuum cleaning robot, it's as if it wants to, you know, vacuum stuff and make a map of the room, but it doesn't literally do it. It just sort of has a structure that's sufficient for us to go, oh, it's as if it maps the room, but it doesn't literally map the room, right?

And of course, Dennett wanted to either say that all intentionality is all, is as if intentionality, and that as if intentionality meant adopting the intentional stance and interpreting an entity as if it had intentionality.

So then if you apply that to yourself, what is your interpreting activity?

Is it intentional or not?

So Dennett, that's the kind of the incoherent.

I'm not as if.

I mean, I know I'm not as if.

Yes.

Oh, of course, Dennett would say, oh, no, you don't know that.

Right?

So, yeah.

Yeah.

Those, I mean, those questions are very important questions.

And I would go back even further than John Searle to Leibniz, the Leibniz mill, the cogs inside the mill.

Do they know anything about the wind sort of thing?

Sure.

But John Searle made the Chinese room argument
that said computers can't have meaning inside them.

And to my mind,

nobody produced a satisfactory answer.

Yeah, I agree with you on that.

I've always thought that.

I think, like,

there was a lot of debate about it.

It lasted to about the mid-90s,

and then people just quit talking about it.

Well, also, David Chalmers.

Yeah.

I mean, I think David Chalmers came back to it
in about the 2000s.

He said computation is a causal network of causal topology,
and that has meaning.

I think his answer was unsatisfactory,
but we could talk about that.

But I think, basically,

nobody's really answered John Searle.

And the conclusion that I'm reaching

is that it agrees with John Searle

that you need something else besides computers
going on to get consciousness.

I see that.

And then there's also, you know,
those old sort of famous results
in mathematical logic.

I mean, other than Godel's incompleteness theorems.

I'm talking about the Löwenheim-Skolem theorems.

Gosh.

Where you can have, you know,
non-standard models of domains
where it's like just a first-order, you know,
model of even arithmetic
doesn't really capture arithmetic.

Yeah, sure.

No, I'm...

It requires a kind of,
oh, we're going to interpret it as just being this.
And then I may have gotten that one wrong,
but there are results like that.

And then there's the whole issue
of the non-categoricity of first-order models.

Yeah.

So it's like second-order.

But we don't have a good idea
of how to compute second-order models
to get categoricity.

No, no.

My views on mathematics are unpopular.

I would say mathematics...

I mean, everybody sees mathematics
as this separate abstract domain.

Actually, I think
certain elementary facts of mathematics,
like 2 plus 2 equals 4,
they're about numbers.

They're not abstract.

And they are experimental facts.

And they agree with the data.

When you have two things
and add two things,
you get four things.

And we've learned that.

Yeah.

But I thought, you know,
in terms...

The only alternative
a computationalist has here
is to say something like
there are objective facts
that pertain to, say,
what a computation...
what a realized computation computes

that are kind of intrinsic to it.

And that's almost...

This is exactly...

Yeah, that's almost

a kind of Pythagorean position

you would have to hold.

Well, I think that is

straightforwardly wrong.

In other words,

I can have a computation

like a logical AND gate.

I can have two 1s make a 1.

But I can interpret that

either as an AND gate

or if I interpret...

If I change around and say,

no, 1 means true

and false and 0 means true,

I can say,

no, it's not an AND gate.

You're not doing AND,

it's doing OR.

Or again,

in terms of bitwise operations,

I can say,

this operation on bits

in the computer

is an addition.

Or I can say,

no,

I haven't represented...

used bits to represent

1, 2, 3, 4.

I've used bits to represent

1, 10, 100, 1,000.

So the operation

it's actually doing

is multiplication.

And so there is no objective fact
about what operation,
what computation is doing.

Yeah.

I mean,
if you think that,
then I think you're right.

So I've often thought this,
that what the computationalist
has to say
is that,

no,
there are objective facts
about what
certain realized
computational structures
compute.

And maybe,
was Carl saying
something like that
before he got up
about the free energy?

I think he was,
yeah.

Yeah.

It's a shame
because I've got
slides coming up
about the free energy principle
and Markov blankets
and so on.

We could have discussed them then.

But anyway,
I'll get to discuss that
with Carl later
because I'm giving this talk
to the TNB later.

So,

but I think Carl
was sort of saying
that there is an interpretation,
a unique interpretation
of a computation.

And I think
there are just too many
counterexamples.

I think
any computation
you show me,
I can,
and you show me
your interpretation of it,
I can show you
another interpretation
that's just as good.

Yeah.

Yeah.

Well,
when you said,
if I may say one last thing,
when you,
when you talked about,
and David mentioned this,
the sensory
and motor
encoding
as a kind of
locus
of real
representational
content,
is that how I,
yeah.

Sensory,
sensory organs
encode

and muscles

decode.

Okay,

why couldn't somebody

raise the same

objection

that you're raising

to computation

as

encoding,

meaning

to that process?

Because there again,

you might say,

well,

it depends,

you know,

to kind of look

at another

line of

theory of

intentionality,

theory of meaning,

theory of representation

that went on

around the same time,

there's,

for example,

Ruth Milliken's

sort of naturalistic

theory

that would allow you

to assign

a definite content

to sensory

and motor,

you know,

encoding

and decoding,
but it's all
kind of based
on evolutionary
history.
And if the
evolutionary history
were different,
but you had
exactly the same
physical inputs,
outputs,
and so on,
you would get
a different content.
And that's
kind of
an unsatisfaction,
to my way of thinking,
kind of unsatisfaction,
for,
it may be fine
with respect
to,
like,
what the body
is objectively
doing and what
it evolved for,
but in terms
of conscious content,
it seems to me
it's not going
to work.
It's too
externalist.
It makes it,
like,

even if I had
a completely
different evolution
and,
you know,
but I ended up
being this kind
of atom-for-atom
replica
of what I am now,
the contents
of my thoughts
don't really change,
it seems to me.

Well,
yeah,
I mean,
I would say
you can't throw
evolution away
like that,
or you can't
play with it
like that.

I think the brain
is a result
of evolution,
and evolution
has meant
that sense organs
have evolved
to encode
light and touch
and so on,
and muscles
have evolved
to do things,
and brains

have evolved
to make sure
that they do
the right things,
and all that
is a coherent
picture,
and I cannot
just pull that
apart,
basically.
So you,
if I understand
you,
you might,
you,
would you think
that you can't
play the same
objection
to the
representational
content
of sensory,
sensory encoding
and motor
decoding,
or however you
want to think
of it?
Well,
I think,
as you can play
a step back
a bit,
I think all
the theories
we have

in natural
science
or gas laws
or whatever
it might be
involve
encoding
and decoding,
and the
theory we have
of the brain
involves encoding
by sense organs,
involves decoding
by muscles,
so that is,
that theory
is a coherent
package
that we can
test against data
on its own,
whereas the
difficulty with
consciousness
is that
we don't
see the
decoding,
whereas we
know that
consciousness
is about
decoded
information,
but ordinary
physical theories
or theories

of the brain,
they always
have an
encoding bit
and a
decoding bit
and they
kind of
balance out,
so we've
got,
at the end
of the day,
predictions
we can test
against data
and we can
say this
theory is
right or
wrong,
so that's
roughly my
viewpoint,
but there's
a lot there,
we're getting
into deep
water.
Yeah.
Thanks.
Robert,
we've also,
of course,
emailed on
this,
but external
information is

required to
interpret those
physical events
as a
computation,
so I
think I'm
100% with
you on
relational
semiosis
and the
idea that
multiple
perspectives
could be
taken on
a
computational
process,
but what
causal
relevance
other than
actually
intervening
on the
internal
states
would those
external
states
have.
So those
interpretations
matter for
the
interpreter,

but unless
they re-enter
causally into
the target
system,
how do they
come to
matter for
the system
itself?

the

So,

I think that's
what I'm
going to
address on
the second
half of
the talk.

And again,
the whole
picture is
the computer
only picture
is radically
incomplete,
and you get
gaps like
that,
but there's
something else
comes along
which fills
the gap,
and that's
what I'm
going to
talk about.

Okay,
so,
that,
by the
way,
we can
come back
to that,
but that's
more discussion.

I thought
some people
might object,
might have
that question,
which in a
sense was
Carl's
question,
that isn't
every physical
thing a
computation?

And to a
certain extent,
I can regard
lots of
physical phenomena
as computation,
I can regard
a gas in
a cylinder
as a
computation,
and there
the encoding
is something
that gives

the gas
a certain
temperature,
but my
theory is
not just
gas in
a cylinder,
it's gas
plus a
pressure piston
which measures
the pressure,
and the
pressure piston
is what
does the
decoding.

So,
that's a bit
of a diversion.
I'd like to
move on from
this slide,
because that's
not what
the second
half of the
talk is about.

If anyone
wants to come
back to this
slide later,
I'm happy
to do so.

Now,
what's the
positive

proposal?

How can
brains be
conscious?

The
positive
proposal is
the brain
is a neural
computer,
but it's
not only
that.

Something
else is
happening,
and it
has to be
happening,
because we
know brains
are conscious,
and so we
need something
else to make
them conscious.

So,
what might
that something
be?

the proposal
here is
that the
extra
something is
an analog
model of
reality,

and that
an analog
model is
different from
what goes on
in digital
computers or
neural computers.

So,
what's an
analog
model?

Approach this
by examples.

A toy car
is an analog
model of a
real car,
or a map
is an analog
model of a
piece of
terrain,
scene,
or a
photograph
is an
analog
model of
some
scene.

So,
this gives
us some
idea of
what
analog
models are

about,
and I
claim we
try and
encapsulate this
in the definition
of what is an
analog model.

A model M
is an
analog model of
some physical
situation P ,
if M has
the same
topology as
 P .

In other
words,
if the
situation P
is
donut-shaped,
then M
is donut-shaped,
and there
is some
kind of
point-to-point
correspondence
between the
physical situation
 P and the
model M .

So,
a point in
the model
is a

function of
a point in
the physical
thing,
and the
function F ,
which relates
points in
model and
physical reality,
are a
continuous
function,
mathematically
it's a
fiber bundle,
I think,
and it
preserved
neighborhoods,
and it
preserved
spatial
binding,
so there
are things
in the
analog model
which are
like borders,
for instance,
borders on
a map,
which are
also borders
in the
real situation,
so some

of the
information
in the
model M,
in the
physical
situation P,
is reproduced
in the
model M,
but with
no encoding,
and so
there's certain
information you
can get out
of the
model M,
which doesn't
require decoding,
and that is,
I believe,
the secret to
consciousness,
that there is
an analog model
inside the
brain,
and that
model is a
source of
consciousness,
that model
doesn't need
any decoding,
so there can
be information
in consciousness

about external
things,
because it's
gone into the
analog model
without being
encoded,
so,
I should say
that I've
tried to
discuss
analog models
in a couple
of minutes,
but they're
not that
simple,
because you
could say
analog computation
requires external
decoding,
just like any
other form of
computation,
and that's
what I believe
also has
said,
and there's
been loads
of philosophical
debate about
what analog
modeling is,
people like
Gibson,

and so on,
and you
could say,
well,
you've given
the examples
maps and
photographs,
but they're
actually encoded,
they are
divisible,
you look very
closely at a
map,
you can divide
it into
little fibers
on the paper
and they've
got ink,
or they haven't
got ink,
and the ink
is an encoding
of information,
or you look
at a photographic
film and there
are molecules
of dye,
or you look
at a computer
screen,
there are pixels,
they're all
encoded,
however,

those aren't
the only
examples of
analog models,
there are
other analog
models which
aren't divisible,
for instance,
if you have
a lens,
the optical
image is a
model of
reality,
but it is
not divisible,
because you
zoom in very
closely on the
image,
you get to
the diffraction
limit,
and usually
get blurry,
so you cannot
keep dividing
an image from
a lens as
far as you
like,
similarly,
a hologram,
it's not
divisible,
because a
hologram is a

Fourier transform
of a spatial
distribution on
a film,
say,
and so you
can't zoom in
as close as
you like on
a hologram and
keep dividing
and dividing it,
and another
more exotic
example is a
single quantum
state,
that's not
divisible,
notoriously not
divisible,
and there are
some quantum
states which can
hold very large
amounts of
information,
macroscopic
amounts of
information,
and they
last longer,
they last a
long time,
most quantum
states get
dissipated in
10 to

minus 18
seconds and
don't last
at all,
but certain
quantum states
can last
very long
times,
and these
are bosons
and condensates,
they're things
like superconductors
or superfluids,
by the way,
Fran is watching
this talk,
I'm running
out of water,
so if she
could bring me
some water,
I'd be very
grateful,
so bosons
and condensates
are an
example of
quantum states
which can
exist for a
long time and
can hold a
lot of
information,
but they
are very

rare states
of matter,
so I
think people
say that
couldn't exist
in the brain,
but I think
we shouldn't
rule them
out because
evolution is
very clever,
so the
question of
analog
representations
in the
brain is
still, I
think,
an open
question,
and it's
a question
of
physics
or
biophysics
rather than
a
philosophical
mathematical
question,
right,
so what's
this got
to do

with
consciousness?

Analog
models have
spatial
binding,
in other
words,
they have
things like
boundaries
which map
to boundaries
in the
real world,
now our
conscious
experience
includes a
3D model
of local
space,
that's what
it's like
to be,
you see a
model of
your screen
in front
of you,
model of
your table
itself,
and that
model has
spatial,
oh my
wolf's

coming,
that's
great,
thank you,
that model
has spatial
binding,
and so
that model
could be
the source
of consciousness
with no
decoding
required,
so here
is the
positive
proposal,
the brain
as well
as a
computer
contains
an
analog
model
of
local
space,
and that
analog
model
is the
sole
source
of
consciousness,

and consciousness

is nothing

to do with

the neurons

in the

brain,

it's all

coming from

that

analog

model,

so that's

the positive

proposal,

and I'll

sketch a

bit,

so the

question you

might ask

is,

well you've

said there

might be an

analog

model of

space in

the brain,

what use

would that

be,

why would

it evolve,

would it

serve any

purpose,

would it

help the

brain do
any real
computing
brains need
to do,
and the
answer is
yes it
could be
very useful
because it
could store
a Bayesian
maximum likelihood
model of
what's going
on around
the animal
for limited
periods of
time,
say for a
few seconds,
it could be
the place
where all
the sense
data about
those animal
surroundings
get integrated
into a single
best likelihood
model,
and it could
persist this
information for
a fraction of

a second while
the animal
decides what
to do,
it could keep
this information
while the
animal classifies
the things
around it,
recognising
things that
they have to
do things
about,
it could be
used for
simulating and
planning what
the animal's
going to do,
and it could
be used for
comparing the
present situation
with past
situations like
memory and
learning.
so the
answer is
yes,
there could
be lots
of uses
for an
analogue
model in

the brain.

Could it

happen?

There is a

possible physical

form for the

model as a

wave excitation

and I've

written papers

about this,

but I'm not

going to

divert into

those papers

now because

that is

simply,

for the

purpose of

this talk,

just an

existent

proof,

to prove

that it

can be

done,

it could

be done,

if evolution

could use a

wave excitation

in the

brain to

store

information,

then that

could be
where the
analogue
model is
stored.

And just
to illustrate
that in
terms of
free energy
principle and
active
inference,

here is a
picture of
the classic
active
inference
FEP

model of
the brain.

The brain
is a
neural
computer and

it has a
Markov
blanket,
there's

an outer
Markov
blanket
between the
brain and
the rest
of the
world.

And the

sense organs
are outside
the Markov
blanket,
they encode
information
which comes
through the
Markov
blanket,
the brain
does
computation
and then
neural
spike trains
come out
of the
Markov
blanket and
they go
to muscles
which turn
into physical
actions.
So there
is encoding
and decoding
in this
computer model
of the
brain,
but there's
no decoding
inside the
brain.
so active
inference is

all about
this cycle
here where
the muscle
actions produce
all sorts of
changes in
the world
and they
come back
by a
sense
organ.

So active
inference
and the
FEP is a
very complete
and capable
model of
the brain,
but I
would claim
it cannot
give you
consciousness
because all
the information
in here is
encoded and
there's nothing
inside the
brain that
does the
decoding.

Decoding is
done on the
way out.

So how
would active
inference and
the FEP have
to be modified
to take
account of
consciousness?

It's as in
this diagram
here, you
have not only
an outer
Markov, you
have the
same senses
and muscles
in the
outer Markov
blanket as
you had
before, but
also as
well as the
computer in
the brain, the
gray computer
we've got
here, we've
got the
inner ring,
the inner
right, white
ring, which
is the
analog model
of local
space.

So what happens is sense data comes in, has encoded the brain, and then it gets decoded again into this analog model.

And from the analog model it gets encoded and there's a rapid iteration of comparing sense data with the model.

So that's the way the best possible model of local space gets developed inside the brain.

and similarly if the animal wants to plan things, then it extracts information from the analog model about the current

situation, it
does some
computation about
it, and it
puts back
what the
situation would
be if the
animal made
certain moves.
So if the
animal wants
to plan
certain movements,
this green
cycle here is
what happens.
and so in
this theory of
enhanced
accurate inference,
if you like,
there is an
internal analog
model as well
as the
computer, and
consciousness
arises solely
from that
internal analog
model.
The advantage
of that is
that if you
get consciousness
from the
analog model,

you don't
need to
decode it.
And the
analog model
is just
like what
consciousness
is just
like, so
there's no
decoding
involved, and
we've solved
the problem
which we
had with
purely neural
computer
models of
the brain.
Okay, so
that I
think is a
future
direction which
I would hope
active inference
would take
in order to
produce a
decent account
of consciousness,
and I
think that's
a promising
route.
So, just

to talk
briefly about
the future
for theories
of consciousness,
I think some
approaches, most
approaches to
theories of
consciousness
have this
brain as a
computer-only
assumption, and
therefore they
can't actually
account for
consciousness
because they
don't have
decoding.
But I think
some of the
theories do
have some
promising elements
in them.
I think the
global
workspace
theory,
there,
it's
compatible
with the
idea that
the global
workspace

could contain
an
analog
model of
space
somewhere in
the
thalamocortical
system.

Certain
people have
written,
Bjorn
Merker has
written about
consciousness
without
a
course.

Again, this
is compatible
with the
idea that
somewhere in
the
midbrain,
possibly
the
thalamus,
there is an
analog
model.

I'll talk
briefly on the
projective
consciousness
model, which
some of you

know very
well, and
that says
there is a
model of
space in
the brain,
which is a
projective
transform
model.

Although the
PCM doesn't
talk about
analog at all,
it does talk
of projective
transform
model.

It has
many
properties in
common with
our consciousness,
and I think
that is very
much a step
in the right
direction.

I've got a
more detail
working out
of this,
which I call
the projective
wave model,
that says
there is an

analog of
space in
the brain,
it's in the
thalamus,
it's in a
wave excitation,
or in
insects,
it's in the
central body,
and that's
the basis
of consciousness.

But that
is not really
the main
point of this
talk,
as you talk
about the
future,
is to
say we
need to
go in
some
different
direction,
because the
ordinary
computational
theories of
consciousness
are not
going to
hack it.
So that

is really
the end
of what
I have
to say
after
35,
how long?
No,
an hour.
So open
for discussion,
here are a
few questions.
Is there
any alternative
to having
an analog
model in the
brain?
Is there
some other
thing that
could be
the cause
of
consciousness?
Could it
be stored
in neurons,
or does it
have to be
stored in
some other
form,
such as a
wave?
And I

think this
has not
only implications
for consciousness,
but also
implications for
cognitive
models of the
brain.

If we
admit this
possibility,
does that
actually help
us produce
models of how
the brain
works?

Or
alternatively,
should we
just give
up on
theories of
consciousness?

So open
for questions.

May I jump
in?

Sure.

Let's go.

So one
thing,
just a second,
I'm going to
close my
door.

All right.

So one
thing to
bear in
mind with
respect to
this
discussion
about the
brain in
particular,
is that
for most
neurophysiologists,
brain
scientists,
not cognitive
brain
scientists,
as you
know,
who often
so are
not real
neuroscientists,
have said
and restated
when,
after computers
in their
digital
form,
so with
this encoding
and decoding
process involved
have emerged,
that the
brain is not

a computer
in the sense
that precisely
the brain
is an
analog
machine,
that neurons
do not
encode things
that actually
spike strains
are analog
signals.

So it's
actually the
core belief
of the
classical
neurophysiology
and computer,
sorry,
brain
scientists,
that the
brain is
not digital
but an
analog
machine
that has
analog
model of
the world.

Now in
the detail
how that
would work

and how
that would
generate
phenomenology
as the
subjective
experience
we have
is,
I guess,
another
question.
So I
think that
what you
are stating
is actually
very
classical
from the
perspective
of
neuroscience
but it
has been
forgotten
by a lot
of people
who are
in the
business
of
cognitive
neuroscience
and
myself
at some
point

I
thought
we have
to go
back
to the
analogy
between
mind and
brain
and software
and hardware
so
entailing
some sort
of encoding
decoding
through
virtualization
but that's
the theory
of mind
that I
don't know
what I believe
today.
but actually
what you
say is
about
can the
brain
be
considered
or part
of the
brain
or neurons

can encode

analog

model.

For most

neurophysiologists

that's already

the case.

How is

another

question?

So now

in terms

of the

particular

model

and

implementation

or form

of this

analog

model

I would

even

go further

and say

at this

point in

my career

and reflection

where I'm

actually

completely

lost

I'm

open to

anything

including

what I

used to
consider
as
sort of
science
like the
claims of
the Chalmers
about quantum
mechanics
why not
and even
some other
physics that
we don't even
know yet
and haven't
understood
in terms of
the brain
maybe
I'm going
to use
an analogy
like an
antenna
of source
receiver
and emitter
that connect
to physics
in which
consciousness
resides
we may
never
understood
so

the field
is open
for me
and
I'm
not shocked
by your
arguments
I think
it's worthwhile
to pursue
but I don't
have a definite
answer
neither
of
what it could
be
this analog
model
or even
if there is
not a physics
in which
encoding
and decoding
can perfectly
make sense
okay
I've got
two remarks
I suppose
I mean
I
well
let me
remember
one of

the idea
that the
brain is
an analog
computer
and neuron
spike trains
are an
analog
representation
of reality
I don't
think that
to me
spike trains
are
irretrievably
encoded
and we
don't know
how they're
encoded
we don't
know
certain neurons
pyramidal neurons
maybe firing rate
is the way
to look at it
but even
looking at firing
rates
is decoding
but nobody
in neurophysiology
looks at
firing rate
in a

computational
sense of the
term
as computational
neuroscientists
do
they just
consider
it's an
analog
signal
that
has an
analog
relationship
with the
amount
of neurotransmitters
that are
going to be
released
and how
that's
going to
affect
based on
the
biochemistry
of the
synaptic
cleft
the potential
of membrane
on other
neurons
so for them
it's really
like an

electronic
machine
of a
classical
analog
you know
there is
no analog
digital
converter
if you
wish
in the
sense
of
analog
signal
for them
it's
all
analog
I suppose
the other
problem
I have
with that
is you
can look
at certain
neurons
in the
brain
and it
appears
that they
encode
things
or they

represent
things
by firing
rate
you could
say
that is
analog
but actually
how do you
know
models

I'm sorry
to cut
you
Robert
the
enthusiasm
of the
discussion
but those
are models
that have
been
extracted
and designed
by people
who started
thinking
computationally
about the
brain
and who
had a
fairly
superficial
culture

about neurons
and
electrophysiology
I would say
for many
of them
and that
decided to
interpret
spike
trains
and their
frequency
or the
rate of
spiking
as a
signal
that in
itself
was the
meaningful
signals
in their
simplified
model
of
neurophysiology
to be
able to
compute
certain
functions
like
doing
arithmetics
or
understand

certain
basic
memory
mechanism
and input
output
relationships
through
transfer
function
but that's
a small
very small
part of
neuroscience
that's
the
computational
neuroscientists
that were
trying to
do
modeling
and that
interpreted
spike
train
as a
relevant
parameter
but a
true
neurophysiologist
would never
think that
way
do you mean
experimental

neurophysiologist
or even
theoretical
I mean
an
electrophysiologist
who masters
the biophysics
of
membranes
of active
conduction
oh sure
that's a
physical
theory
that way
they think
that
yeah
I'm talking
about people
who worry
about consciousness
yeah but
they are not
they are usually
not real brain
scientists
that's what
I'm saying
yeah
okay
Kenneth
and then
Paul or
anyone else
who hasn't

spoken

Ben

unmute

Kenneth

sorry

just to

make a

quick remark

with

relation

to what

David

just said

I mean

I think

McCulloch

and

Pitts

knew their

neurophysiology

pretty well

and they

knew they

were making

a massive

simplification

and

making

a choice

and kind

of setting

up neural

network

modeling

you know

as computational

but

I just

want to
say
and it's
also kind
of following
up with
what
David
says
well I
have two
questions
really
so one
is
suppose there
is this
analog
model
I mean
can't we
raise the
same sort
of objections
to computational
models
that we
I mean
couldn't we
raise to
analog models
the same
sort of
objections
we're raising
to computational
models
models

so for
example
if you
take
analog
computers
like
the
differential
engine
you know
Charles Babbage's
wasn't it
called the
differential
engine
that
yeah
you know
that was
Latin Babbage
well
he did
the differential
engine
and there
was a
laser one
in the
30s
okay
so it's
it's using
gears
if I'm
not mistaken
to do
yeah

sure
you mentioned
the abacus
I mean
it's an
analog
device
and someone
might kind
of push
and go
well
really
digital
computers
are all
implemented
in an
analog
world
so really
they're
analog
I know
there's like
the flip
side
where everybody's
trying to
turn everything
and everything
analog into
something digital
like you know
information
ontologies
or it
it from

bit
kind of
so there's
this kind
of
to me
the key
distinguishing
factor is
that some
of the
information
in an
analog
model
doesn't
need
decoding
okay
and I
can
I can
I can
I like
that idea
because I
think that
there's
and that
leads me
to my
second
question
I like
the idea
because
I'm one
of those

who thinks
that
consciousness
is not
exhausted
by intentionality
and what I
mean by that
is
yes there's
the intentional
level where
you think
like you're
seeing something
as
something
but there's
also the
level at
which you're
just seeing
if we're
talking about
for example
so for
example
if you
you might
wake up
in the
morning
and have
like a
wadded
sheet
in front
of your

face
and for
a second
you don't
know what
it is
but there's
still
before you
interpret
it
right
so there's
still something
you're
sort of
directly
aware
of
that
doesn't
require
any
decoding
to be
aware
of it
and then
there's
this
further
layer
of decoding
which is
like oh
it's a
sheet
or oh

it's kind
of perceptual
and that's
where
intentionality
or meaning
in the sense
we've been
talking about
it
enters in
now
okay so
I like
that idea
that an
analog model
might give
you that
layer
that isn't
it's
information
that doesn't
have to be
decoded
because it's
really sort
of just
built into
the structure
of that
episode
or that
phase of
consciousness
itself
but

my question
for you
really is
what is it
to decode
or interpret
or intentionally
animate
to use a
Husserlian
language
like what
is it
to see
something
as
something
that's
like a
further
process
you can
imagine
in theory
anyway
a kind
of consciousness
that doesn't
do it
that is
basically
just
kind of
locked
in this
you know
it doesn't
really sort

of get
outside
of itself
yeah
your arguments
earlier on
kind of
depend on
consciousness
always involving
intentionality
or interpretation
which
you know
I certainly
a tempting
thought
that it
always does
I can't
say that
it's
false
but I
wonder
like
what
is
the
theory
of
interpretation
or the
theory
of
intentionality
and do
you need

one
for your
project
well
this
I would
say
in that
moment
you
wake up
before
you've
interpreted
your
sense
data
as
this
is
a
sheet
there
are
things
in
your
sense
data
like
a
boundary
between
red
and
green
yeah
you know

and
you're
aware
of
that
boundary
before
you
say
oh
that's
about
my
sheet
yeah
and so
there
is
a
pre
interpretation
experience
raw
experience
and that
quality
of the
what it
is like
bit
is
what
where the
analog
model
represents
it
has

a
nice
fit
to
consciousness
analog
model
has got
a
barrier
between
red
and
green
your
consciousness
has
that
and
that's
for
classification
and
intentionality
when you
see
something
on the
wall
you
say
oh
that's
a
fly
that
as
built

in
by
evolution
animals
have
to
take
their
sense
data
and
classify
it
into
things
they
need
to
do
things
about
as
a
fly
I
can
eat
it
whatever
and
that
secondary
experience
of
taking
a piece
of
sense

data
and
classifying
it
as
something
is
useful
for
the
animal
but
it
also
involves
the
conscious
bit
that
once
the
animal
has
seen
a fly
as
a
fly
doesn't
quite
see
the
raw
sense
data
it
mixes
the

fly
up
with
previous
flies
it
seen
in
his
memory
and
its
current
model
of
reality
involves
not only
the
sense
data
coming
in
but
all
the
flies
it
seen
before
that
they're
going
to
behave
that
way
and

be
edible
or
whatever
it
may
be
I
can
agree
with
that
but
what
I
wonder
is
is
that
the
seeing
as
or
the
interpretation
you want
to deny
that
that
could
be
computational
correct
no
no
it's
computational
but it

feeds results
back into
the model
so you
think
that
when I
see a
fly on
the wall
there's
computation
in my
brain
that says
that's
a fly
but then
that
computation
feeds back
into my
analog
model
so my
analog
model
is a
bit more
fly-like
than it
was
before
but now
you've
got
intentionality
in your

computations
and then
we can
go back
and go
well
why were
you
objecting
to
computations
being
intentional
in the
first half
of your
talk
no
my
computation
has
evolved
to
classify
the
visual
field
into
useful
things
that I
need
to
know
about
and
you
know

the
computation
doesn't
know
that's
a
fly
the
computation
just
does
some
grinding
and
it
makes
my
muscles
do
a
fly
thing
and it
makes
my
conscious
awareness
do
a
fly
thing
but
inside
it
the
computation
is not
aware

of a
fly
oh
okay
I
all right
I'll have to
think about it
I'm not
I'm not sure
I follow you
completely
but
well
my
brain's
evolved
to
classify
things
usefully
to help
me
survive
I
agree
with
that
yeah
and
that
evolution
hasn't
made it
aware
of what
things
are

it's
just
for
certain
inputs
it
produces
good
outputs
which
make
me
avoid
flies
okay
but
also
produces
outputs
into
the
analog
model
that
says
when I
see
a
blur
on
the
wall
and
I'm
not
sure
what
it

is
I
suddenly
interpret
that
blur
becomes
a bit
more
focused
as a
fly
because
I
know
there
are
flies
on
that
wall
okay
so
the
interpretation
is a
consciousness
is a
conscious
matter
yeah
no
the
interpretation
happens
in the
computer
bit

but
the
result
of
being
more
fly
like
than
a
blur
on
the
wall
is
in
the
conscious
bit
then
it
seems
to
me
you're
using
interpretation
or decoding
in
one sense
in the
first half
your talk
and in
another
sense
right now
I don't

understand
I'm
saying
the decoding
happens
on
in
the brain
has
encode
the computer
brain
has encoded
stuff in
it
and there's
two kinds
of decoding
there's decoding
going out
the muscles
and there's
decoding
going into
the
analog
model
at some
point
if you
wish
Ken
for what
I
understand
there is
a decoding
that comes

from the
processing
of the
within
the
brain
that
involved
initially
a form
of
encoding
that
remaps
somehow
this
code
into
an
analog
model
that
maps
the
outside
world
in
a
manner
that
is
coherent
in
terms
of
shape
if
you

wish
so
there's
a
shape
outside
the
world
you
don't
see
it
directly
you
encode
it
through
neural
process
and
somewhere
somehow
through
some
mechanisms
to be
defined
it's
decoded
within
to
generate
and
synthesize
an
analog
experience
of the

world
for which
even though
there is
an
indirect
process
there is
a
one-to-one
of
sort
mapping
or
some
mapping
that
is
more
analogous
in
the
sense
of
continuous
deformation
if
I
understand
it
yeah
but
this
is
the same
kind
of
decoding

that
can
happen
when you
ask
chat
GPT
a question
and it
spits out
an answer
right
no it
just does
electronics
what's that
chat GPT
just does
electronics
nothing else
yeah I got you
but we're talking
about the
you want to
say there is
still a
computational
sense of
encoding
and decoding
yes
but earlier
in the talk
it was
about
there's
really
no

computation
without
some
external
observer
sort
of
attributing
or
interpreting
a
computation
right
I'm totally
fine
with your
model
and that's
the role
of the
analog
model
if I
understand
well
yeah
yeah
yeah
so there's
a kind
of
there's
sensory
inputs
they get
encoded
and then
there's a

decoding
step
from the
sensory
inputs
to the
analog
model
and when
that's right
and the
analog
model
is like
the
conscious
episode
and it's
what we
might call
the
conscious
interpretation
that's
right
yeah
I
go back
to this
diagram
if
if you
like
I've
just
come back
to
um

is that
diagram
showing
yeah
but earlier
you said
decoding
was going
to be
the same
as
interpretation
and so
what I'm
getting at
is like
the d
like
for your
your
your
your
your
outer
d
to
muscles
is that
really
interpretation
no but
again
I think
from the
they are
decoding
spike trains
they are

interpreting
spike trains
as physical
movements
yeah
from the
outset
Robert
said
if you
only take
the
computer
model
which
in the
digital
world
excludes
an
analog
model
in this
rationale
then you
cannot get
an
interpreter
that is
inside
the
brain
it's
always
some
human
or
some

other
type
of
interpreter
outside
now
what
is
introducing
is
a
new
mechanism
that
is
still
to
be
uncovered
in
terms
of
its
physics
or
its
whatever
in
which
an
analog
model
is
actually
working
within
the
brain

and
therefore
is
not
digitally
operating
as
in
a
computer
so
there
is
a
mixed
architecture
between
a
digital
and
analog
system
in
the
brain
so to
speak
yes
but
but
just
it
just
it
seems
to
me
like

your
D
in
the
red
box
that
process
is not
it's
the
coding
process
is not
itself
an
intentional
interpretation
but it
results
in
one
yeah
it
results
in
one
yeah
it's
creating
the
analog
model
okay
Benjamin
and then
Paul
and then

anyone
else
who
wants
to
add
more
unmute
Benjamin
Benjamin
unmute
apologies
if I
understand
correctly
and I'm
sure you
can clarify
this for
me
there are
according to
this
projective
way
theory
of
consciousness
there are
certain
things
that
are
built
in
or
baked
into

the
system
such
as
time
pressure
or
time
perception
and
are
there
other
things
besides
time
and
space
that
would be
necessary
for
consciousness
to emerge
or
is it
just
that
with
the
effects
of
entropy
I
would
put
the
emphasis

on
space
the
analog
model
is a
model
of
space
in the
present
moment
and I
think
for most
animals
the
present
moment
is all
that
exists
they don't
have a sense
of
chronological
time
and they don't
worry about
time
they're just
worrying about
what they're
going to do
right now
and so
that's
for that

they need
analog
model
of
space
so
I
would
put
much
more
emphasis
on
space
than
time
I
think
chronological
time
is a
recent
human
thing
it
came
along
with
language
great
thank you
Brianna
then Paul
or anyone
else
hi
so
what if

I look at
something
and I
can't
quite
comprehend
it
would
that
mean
I'm
not
conscious
of
I'm
not
currently
conscious
what if
an
animal
can't
comprehend
something
or
anything
for that
matter
sorry
I
didn't
catch
all the
questions
try
again
so
the

ability
to
extract
meaning
in
an
analog
model
is
what
you
would
say
is
required
for
consciousness
done
yeah
so
what if
I'm
unable
to
extract
meaning
about
something
am I
not
conscious
in that
moment
no
I think
you're asking
about animals
I think

we've got
a perfectly
good theory
of animals
how animal
brains
absorb
sense data
and classify
sense data
into situations
they need to do
things about
and do things
about those
situations
that theory
doesn't tell
us how
animal
brains
are
conscious
and
that's
so
I would
decouple
the two
issues
of
you know
how a
brain
does
its
practical
job

of helping
the animal
or the
person
survive
versus
how a
brain
actually
creates
consciousness
now we
don't
actually
know
whether
animals
are
conscious
but
I
would
guess
that
they
are
so
if
they're
unable
to make
the
classification
of for
example
the
sheep
from the

early
analogy
then they
would
not
be
conscious
well
I
think
the
two
questions
are
independent
the
animal
brain
can
make
the
classification
that
it
has
to
but
it
can
do
that
without
being
conscious
of
it
perhaps
Brianna

you can
think
of an
example
that
comes
from
neuropsychology
think
about
a
person
who
gets
dementia
or
some
other
very
debilitating
neurological
disease
and for
whom
at some
point
everything
is a
confused
world
which
is
very
difficult
to
interpret
very
disorienting

very
minimalistic
in
terms
of
the
content
of
the
quantity
of
information
that
a
person
can
integrate
but
nevertheless
the
person
still
is
trapped
into
this
confused
and
disorienting
experience
so there
might be
consciousness
but not
necessarily
very
informative
in the

sense
of
adaptive
coupling
with
the
world
so to
speak
would
you
agree
with that
Robert
I
agree
that
if you
take
this
viewpoint
you
can
bounce
it
off
all
sorts
of
situations
like
that
the
person
who's
terminally
confused
and so

on
and I
think
you
keep
getting
insights
as to
how
the
picture
works
in
those
cases
and
it
takes
to
me
quite
a lot
of
sort
of
reflecting
about
what
this
model
of
the
brain
actually
means
and
how
it

affects
how
we
interpret
various
strange
conditions
like
that
yeah
I
agree
it's
instructive
to do
but
I
don't
have
all
the
answers
to
what
comes
out
of
asking
those
questions
so
the
patient
with
dementia
would
be
less

conscious
than
someone
with
that
is
what
we
would
do
!
not
!
No
a
person
with
dementia
might
still
be
very
conscious
of
the
room
around
them
but
in
an
instantaneous
way
rather
than
like
an
animal

has
no
sense
of
the
future
or
the
past
or
causality
or
whatever
it
could
be
just
conscious
of
sensations
in
the
present
moment
yeah
and
for
me
you
know
what
you
say
is
somewhat
what
would
the

integrated
information
theory
would
claim
that
you
need
to
have
some
sort
of
complexity
and
efficiently
integrated
information
that
is
both
rich
and
synthetic
to
get
a
high
quantity
of
consciousness
but
to me
the
core
of
consciousness
is

not
that
and
I
worked
a lot
with
patients
a long
time
ago
in
neuropsychology
and
I
think
you
can
be
acutely
conscious
while
being
totally
confused
it
can
be
a
very
intense
conscious
experience
to be
totally
confused
it's
not

informative
in the
sense
of
adaptive
coupling
but
that's
so
then
meaning
is
not
required
for
consciousness
or
maybe
a bit
of
meaning
but
at
least
some
and
again
that
begs
the
question
of
what
we
call
meaning
it's a
complicated

question
but at least
the structured
adaptive
highly informative
content of
consciousness
when the
wine is
super
efficient
and
well coupled
with its
environment
might not
be what
really
is the
most fundamental
aspect of
consciousness
it often
goes with
it
it's
very useful
to have
it
but
I remember
patient
is a
very
amnesic
patient
with
enterograd

amnesia
that were
very
bad
or
dementia
patient
that
really
were
confused
but
were
living
very
intense
and
distressing
experience
and
I would
say
that
even
though
they
could
not
necessarily
have
a
very
profound
recursive
thinking
about
their
own

consciousness

and

how

it

is

perceived

by

others

etc

even

though

they

would

have

a

hard

time

to

navigate

even

their

immediate

environment

their

experience

was

intense

I

definitely

agree

with

your

interpretation

fascinating

I

think

that's

all

I
have
to
say
hi
Paul
hi
Robert
I
thought
you'd
be
disappointed
if I
didn't
say
something
so
there
we
go
I'm
also
a little
disappointed
you didn't
credit me
with the
Roomba
example
didn't
I
didn't
what
didn't
credit
me
with

the
Roomba
I
didn't
get
in
that
the
rumor
example
is
dear
Paul
first
of
just
to
say
that
I
am
a
lay
person
in
all
of
the
aspects
of
this
discussion
except
for
the
fact
I
am

a
professional
human
being
and
Robert
and I
often
engage
in
philosophical
and
computational
discussions
around
the
area
this
area
and
not
being a
lay
person
I
can
pose
some
questions
but
I
may
not
understand
your
answer
so
I

am
going
to
say
that
right
the
answer
no
may
I
but
Robert's
requirement
for
encoding
and decoding
relies on
the requirement
for an
extra layer
of
significance
and meaning
to use the
word that
he has
given
it
and
my
understanding
is that
those
requirements
for extra
significance
and meaning

come from
the fact
that we
know
we are
conscious
that
we have
consciousness
so
Robert and
I
in one
of our
discussions
came to
be an
impasse
where
he said
well
I know
I can
take the
rimbar apart
and I can
see how
it works
and it's
clearly just a matter
of physics
and electronics
and so we
have that
evidence
and then we
have the
evidence

that we
are conscious
then he
said
at least
I'm
conscious
do you
not know
that you
are
Paul
and
I
don't
know
I don't
know I'm
conscious
in the
sense
that
there is
a clear
category
difference
between
what I
experience
what I
feel
if you
like
and what
I can
imagine
an
advanced

AI
self-evolving
computer
with
sensors
actuators
etc
to have
I cannot
hand on
heart
say that
I know
where that
dividing line
comes
so I
would just
like to
posit
that there
is no
meaning
beyond
what our
brains
our neural
brains
what their
currency
is
what the
conclusion
of our
neural
brains
is
there is

no
meaning
beyond
that
because
both
the
advanced
Roomba
whatever
or
ourselves
we
attach
adequate
meaning
to
what we
perceive
to
inhabit
the
environment
we're
in
so I
would
question
why there
is a
divide
that requires
encoding and
decoding to
happen
I think
in a nutshell
that's

the essence
of the
argument
which I've
rehearsed with
Robert
before
yeah
let me
try and
approach
that
I mean
I'm
concerned
with doing
science
in order
to do
science
I need
experimental
data
and if
I'm going
to do
science
about
consciousness
I need
experimental
data about
consciousness
and I have
got data
I claim
that tells
me I am

conscious
but I
haven't got
data that
tells me
the
rumba
is
conscious
so
the
question
whether
or not
a rumba
is
conscious
I don't
feel
it's
justified
to assume
it is
conscious
in any
sense
anything
it's
like
to be
a
rumba
I just
don't
see
that's
a
problem

for
my
science
if
it's
solely
trying
to
account
for
consciousness
of
things
with
brains
biological
things
so
in a
sense
the
question
whether
the
rumba
is
conscious
or
not
is
of
less
interest
to
me
just
because
I

don't
have
data
about
it
because
I
think
I
do
have
data
about
me
that's
not
the
challenge
I'm
putting
to
you
Robert
the
challenge
I'm
putting
to you
in essence
is
whether
there
is
a
distinction
between
the
sensory

experience
ability
to
operate
in
its
environment
that
are
and
beyond
advanced
AI
self-evolving
possibly
self-replicating
with improvements
robot
between that
and what
we are
so
again
I'm challenging
the notion
that
humans
have some
categorical
distinction
self-awareness
awareness of
our environment
which
we
call
consciousness
which I

have a
real problem
with that
word
which is
different
to that
of an
advanced
piece of
electronics
and
computing
so
it's
it's
I'm not
challenging
you to
say
whether
a
room
bar
is
conscious
or
not
I'm
challenging
you to
identify
a
distinction
between
the
two
because

if
there
is
no
distinction
between
the
two
the
only
distinction
is
I've
got
data
about
me
I
don't
have
data
as a
room
bar
and
I'm
doing
science
so
I
need
data
but
you
the
data
you've
ascribed

to
yourself
is
consciousness
that
is
distinct
from
machines
so
I
have
data
that
says
there
is
something
that
is
like
to
be
me
this
is
the
Thomas
Nagel
sense
of
consciousness
that
is
what
it
is
like

and
internally
in my
brain
in
in
my
life
there
is
something
that
is
like
to
be
alive
I
don't
have
that
evidence
about
a
room
bar
so
a
room
doesn't
worry
me
because
I'm
trying
to do
science
not

just
philosophy
I'm
using
philosophy
for
science
if
you
like
and
for
my
scientific
purposes
I
need
data
and
the
only
data
on
consciousness
that
I've
got
is
data
on
consciousness
in
people
so
I'm
deliberately
restricting
my

focus
if
you
like
just
to
do
this
and
not
that
but
what
you're
saying
is
that
your
consciousness
is
putting
a degree
of
significance
on
the
objects
which
your
eyes
light on
which
is
greater
than
that
simply
required

to
operate
in
your
environment
I
don't
see
yeah
I
mean
it's
a
remarkable
fact
I
could
be
a
zombie
but
I'm
not
right
well
that's
what
I'm
shouting
in
the
politest
possible
way
yeah
I
mean
it's

a
philosophical
possibility
that
I
am
a
zombie
but
I
haven't
had
first
person
experience
which
says
I'm
not
a
zombie
whereas
a
rumba
could
well
be
a
zombie
and
I
think
it
actually
is
a
zombie
it

doesn't
bother
me
I
just
restrict
my
viewpoint
to
where
I've
got
data
I'm
not
sure
that's
what
Paul
is
talking
about
but
I'm
saying
my
viewpoint
is
restricted
I'm
only
trying
to
do
science
I'm
not
trying

to
evaluate
philosophical
possibilities
I'm
only
trying
to
produce
a
scientific
account
of
consciousness
and
the
only
consciousness
I
know
about
is
what
happens
in
biological
things
such
as
me
yeah
and
I'm
not
making
a
philosophical
point

I'm
simply
saying
that
your
requirement
for
encoding
and
decoding
is
only
there
because
you
think
that
there
is
something
there
is
something
some
attachment
of
meaning
and
significance
required
to
those
things
over
and
above
what
is

simply
required
to
operate
in
an
environment
like
a
room
bar
and
I
don't
see
that
I
don't
see
that
yeah
I
mean
part of
the confusion
in these
debates
is that
we're stuck
with language
and language
is encoded
and it's encoded
very badly
and ambiguously
and we use words
with very
different meanings

and so
we can
construct
all these
constructs
about
consciousness
which go
way beyond
the thing
itself
and which
get us
down
blind
alleys
I think
I think
you know
I have
a problem
with the
term
consciousness
and it's
it's
it's
it's
it's
one of
those words
that we're
very bad
I mean
we imagine
when we use
words
we imagine

they're kind
of precise
and say
what we mean
but they
are so
imprecise
it does
surprise me
that the
scientific
community
would continue
to use
a word
like
consciousness
rather than
breaking
it down
to its
consistent
parts
and banning
the use
of the
word
consciousness
it does
not feel
helpful
that's
been
true
well
I think
Brianna
had a

had a
follow-up
isn't that
right
to Paul's
you are
right
okay
go ahead
so building
on Paul's
point
I'm actually
curious
whether
an AI
model's
need to
navigate
its
environment
through
interacting
with us
and processing
tokens
might lead
to its
own
development
of a
subjective
experience
could it
be that
such
operational
requirements

foster an
emergent
intelligence
that we
simply
can't
grasp
because
we
don't
share
the
same
kind
of
subjectivity
in other
words
if these
models
engage
with the
world
in ways
that
parallel
our
interactions
what
exactly
distinguishes
their
processes
from what
we
recognize
as
human

consciousness

and

does

this

experience

involve

extracted

meaning

I don't

think I

have a

neat

answer

to

that

really

I just

come back

to my

view

that

anything

that

goes

on

inside

an

AI

computer

whether

it's

running

a

robot

or

anything

else

is

not
about
things
and
I
have
consciousness
which
seems
to me
to be
about
things
for
you
you
can't
verify
that I'm
conscious
or
that
nothing
else
is
conscious
so
why
can't
we
verify
that
a
machine
is
also
conscious
can

they
extract
meaning
like
we
do
from
their
own
subjective
experience
and
does
that
fit
into
your
model
if
they
are
conscious
the
consciousness
they have
is not
the
subject
I'm
studying
does
that
help
at all
no
probably
not
that

does
help
actually
it's a
fascinating
conversation
yep
how about
each
person
can
give
sort
of
a
final
round
where
does
this
take
Benjamin
and
then
continue
around
hello
yeah
so
I
can
answer
questions
about
like
artificial
intelligence
and

consciousness

no

I don't

think

artificial

intelligence

can

be

conscious

at least

not

distributed

systems

specifically

the

needing

diversity

complexity

these

kind of

things

would

prevent

that

from

occurring

in

any

real

measurable

way

like

in terms

of

subconscious

or

something

of that

nature

the

like

a

thousand

of a

thing

is

not

diversity

it's

not

complexity

it's

just

one

thing

so

in

that

regard

I

don't

think

it's

possible

in

any

way

for

artificial

intelligence

to

achieve

sentient

self-awareness

consciousness

but

it

definitely

looks like

it

and

I think

that's

pretty

cool

certainly

so

if

it

walks

like

a

duck

and

it

talks

like

a

duck

it's

probably

a

duck

yeah

exactly

David and

Kenneth

well

I'm going to be a bit mischievous

I agree with everybody

and I don't know

at this point

so

everything is open to me
almost
not everything
but
I don't know
yeah
I think that is the last word
actually
Ken
well
I mean
I think it's perfectly
conceivable
that you could
have
an AI
that
passes
every kind of
souped up
Turing test
you want
they become
our best friends
or whatever
and they're
complete zombies
I think that's
perfectly possible
at least
it's conceivable
on
the other hand
I took your argument
Robert
to be something like
look
if all you've got

is computation
so if an AI
no matter how
fancy it is
no matter how
behaviorally
equivalent
it is
to a
conscious being
if all it's got
is computation
going on
it's not
conscious
because we know
that these
computations
do not carry
any meaning
they're not
carrying
intentionality
and consciousness
at least
has the capacity
for intentionality
and meaning
that whole
argument
as I said
earlier on
is going to
depend on
just like
do you think
there is
a reasonable

theory of
intentionality
that could
be applied
to computational
systems
or natural
systems
and that
of course
is an
old
debate
as I said
earlier
that goes
back to
even before
Searle and
Dennett
but Searle and
Dennett
and then people
arguing about
intentionality
and representation
after that
and then
kind of
you're right
that Chalmers
kind of
resuscitates
it but
it's also
like
it's a kind
of a debate

that
went somewhat
silent
I think
because
everybody
realized
they have
no idea
what to
say
and
at the end
of the day
but I think
it's good
that you're
bringing it
back up
and that
you're making
us think
about this
in light
of a new
idea
now that
idea
itself
I want to
ask you
one sort
of last
detail
oh let me
just say
in favor
of the word

keeping the word
consciousness
consciousness
no
it's been
tried
before
to
ban the word
consciousness
from the
philosophy of
mind
this was
a very
formidable
attempt
by none
other than
the nefarious
Martin Heidegger
with the word
Dasein
and then of
course you
just end up
replicating
you know
creating more
ambiguity
and you
you know
so I think
we got to
pick some
word
right
consciousness

is fraud
it has an
interesting
history
but we got
to pick
some word
for it
but okay
let me put
that aside
the thing
about your
model
is this
when you
say that
this
analog
model
of space
has to
be
topologically
equivalent
to
space
ambient
space
is that
right
and
physical
reality
yeah
if you've got a
tension
your model

has a
tension
and there's
a
continuous
function
right
the
function
must be
continuous
does that
mean that
the analog
model of
space
is itself
really just
a space
it is in
fact
itself
a kind
of space
and is it
continuous
and is it
going to
require that
the physical
realization
of it
involves
continuity
well
in my
particular
theory

the analog
model
is a
Fourier
transform
of a
spatial
distribution
which
and
consciousness
somehow
involves a
Fourier
transform
but that
gets
technical
but
nevertheless
Fourier
transforms
are very
deep in
physics
and quantum
mechanics
so I
think
Fourier
transforms
having to
do with
consciousness
is not
a big
stretch
that's a

way of
evading the
question
yeah
okay
because
you might
think
no
whatever
is going
on here
like
when
our
experience
of
space
it
cannot
be
like
actually
realized
in something
continuous
or it
can't
no
I think
a real
a miniature
space
inside the
brain
is not
going to
happen

cannot
and we
know
enough
about
the
brains
and
there's
not
some
miniature
model
of the
world
inside
the
brain
yeah
but I
think
a Fourier
transform
in a
wave
is the
way that
can happen
okay
all right
thank you
very much
thank you to
the live
chatters also
for their
vigorous
commentary

so thank
you all
till next
time
all right
bye many
thanks
thank you

Thank you.